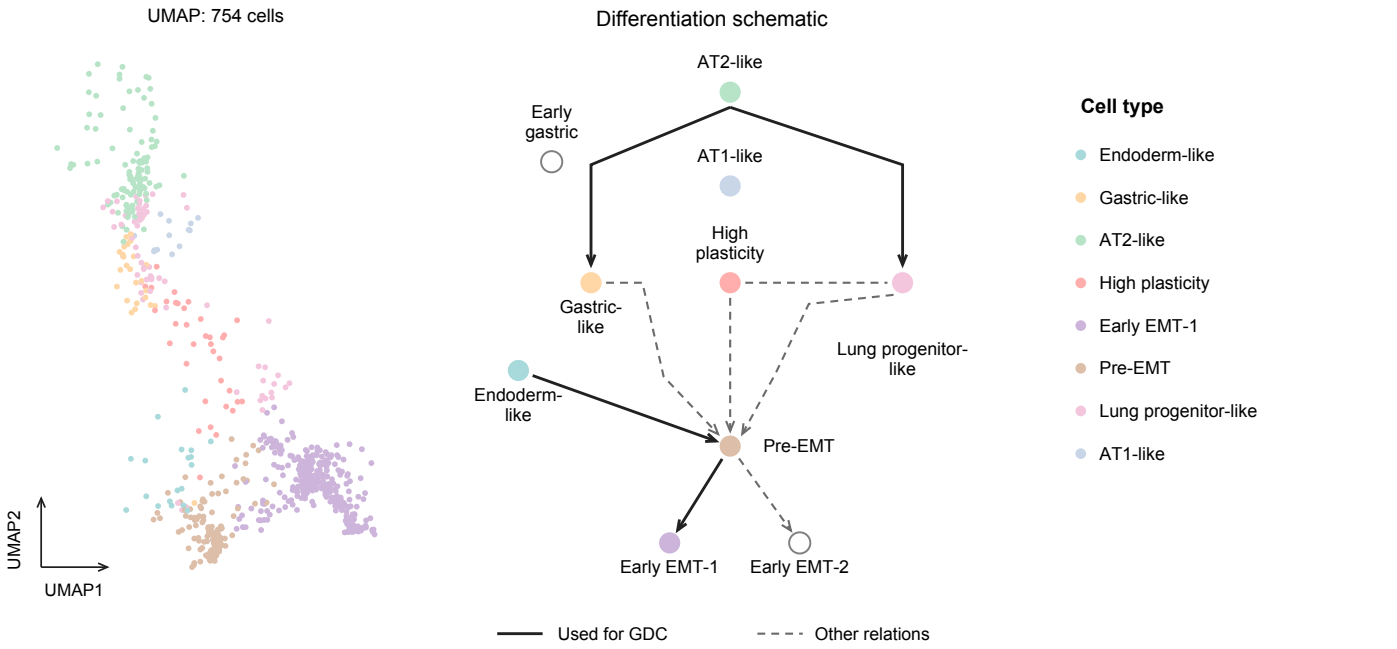
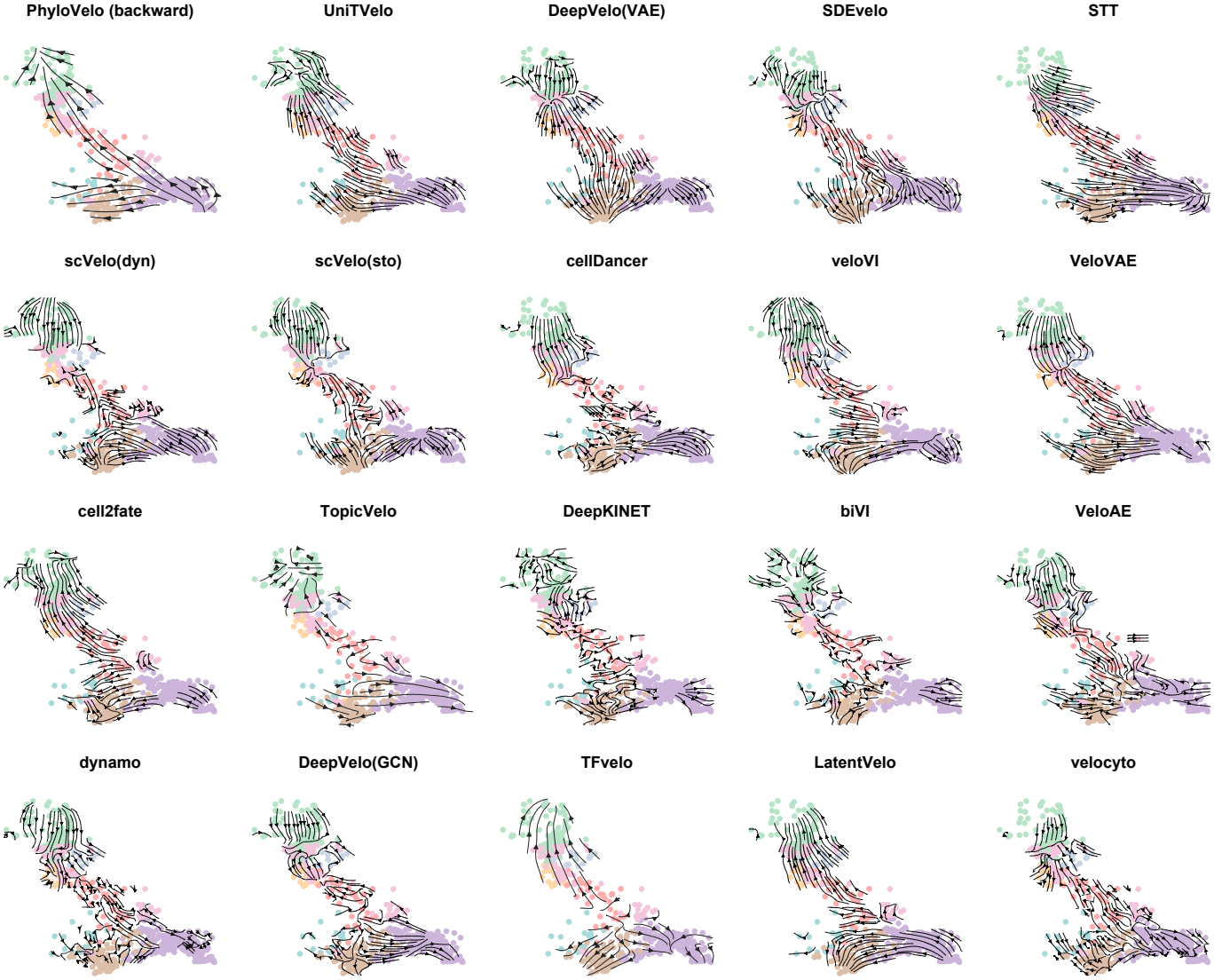
a KPTracer distribution and reference



b Method-specific velocity fields



Extended Data Fig. 11 | KPTracer: reference and method-specific velocity fields. a, All 754 KPTracer cells in the fixed UMAP, with UMAP1/UMAP2 axes and source cell-type colors. The reference schematic is adapted from Yang et al. (Cell 2022, doi:10.1016/j.cell.2022.04.015), Fig. 5I. Solid arrows mark the four relations used for GDC; dashed connections show other literature relations and are excluded from GDC. The dashed horizontal connection denotes an undirected coupling. Line style encodes GDC inclusion, not the original evidence categories. Hollow nodes are literature states absent from this subset. The source Gastric/Endoderm progression is expanded into the evaluated Endoderm relation and a contextual Gastric branch; Early EMT-1/2 is split to identify the evaluated EMT-1 state. Node positions are schematic, not UMAP coordinates or measured trajectories. Lung Mixed labels correspond to Lung progenitor-like. **b,** The 19 existing method fields retain their descending historical GDC order, read left to right and then top to bottom; PhyloVelo (backward) is appended in the twentieth slot. All fields use the fixed cell coordinates; the PhyloVelo field uses the equal-aspect settings of Fig. 5d. TFvelo uses 1,950 finite fitted genes. TopicVelo uses topic 7 of 8 at the 35th-percentile threshold (490 topic cells; 727 nonzero projected directions), retaining the native package fallback. Its gene-space quantity is the native-transition displacement proxy T @ Ms - Ms. Gene sets and preprocessing differ by method. These are descriptive results from one dataset. GDC values are shown in Fig. 5e, with a reference denominator of 706 cells in six participating states. PhyloVelo retains native backward arrows in b; metrics use the negated native projected vectors. Its adapted GDC of +1.000 in Fig. 5e is not a comparable rank because the repeated global gene vector forces within-state coherence to one. This display does not establish PhyloVelo as the best method.